

Systematic *In Silico* Screening and Polypharmacology Profiling of Repurposed Therapeutics Against Eight Hallmark Cancer Targets: Discovery of High-Affinity Microtubule and Transporter Disruptors

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Abstract

Background: De novo oncology drug discovery suffers from protracted developmental timelines and prohibitive failure rates. Drug repurposing capitalizes on established safety and pharmacokinetic profiles. However, systematic, cross-target polypharmacological screens evaluating candidate drugs against distinct functional nodes of oncogenesis under uniform structural conditions remain scarce.

Methods: We conducted a rigorous structure-based virtual screen of 13 approved non-oncology therapeutics spanning anthelmintics, antiparasitics, antifungals, antidiabetics, and anti-inflammatories against 8 hallmark cancer targets: β -tubulin (4O2B), importin- α (4WV6), p21-activated kinase 1 (PAK1, 2HY8), glucose transporter 1 (GLUT1, 5EQI), STAT3 SH2 domain (6NJS), MDM2 (4HG7), Frizzled-4 CRD (6TFB), and thioredoxin reductase 1 (TXNRD1, 2ZZB). Simulations were conducted using AutoDock Vina v1.2.7 with co-crystal pocket centering, RDKit physicochemical parameterization, ligand efficiency (LE) decoupling, atom-level residue contact mapping, and head-to-head benchmarking against standard clinical chemotherapy baselines (colchicine, nocodazole, nutlin-3a, stattic, BAY-876, and staurosporine).

Results: Out of 104 pairs evaluated, 42 pairs (43.8% of organic complexes) demonstrated strong calculated affinity (≤ -7.0 kcal/mol), with 17 pairs achieving top-tier scores (≤ -8.5 kcal/mol). Anthelmintics demonstrated the highest multi-target hit breadth. Crucially, **Praziquantel (PRA)** emerged as an unprecedented high-affinity hit, displaying superior binding to β -tubulin (-10.08 kcal/mol) and GLUT1 (-9.63 kcal/mol) with an industry-leading Ligand Efficiency of 0.438 kcal/mol/heavy atom and optimal Lipophilic Efficiency (LipE = 4.86), significantly surpassing the clinical chemotherapies colchicine (-7.06 kcal/mol) and nocodazole (-8.77 kcal/mol). Residue contact networks verified that praziquantel buries its cyclohexyl core against β -Leu248 and β -Leu255 in the canonical colchicine pocket while forming hydrogen bonds with β -Asn258 (3.01 Å) and β -Lys352 (3.42 Å). **Mebendazole (MEB)** displayed exceptional polypharmacological breadth, binding all 8 oncological targets with ≤ -7.0 kcal/mol (mean affinity -8.09 kcal/mol). **Niclosamide (NIC)** selectively engaged the STAT3 SH2 dimerization cleft (-8.44 kcal/mol), forming four simultaneous hydrogen bonds with the Arg609/Ser611/Ser613/Ser636 triad.

Significance: This study establishes a structural and thermodynamic basis for the antineoplastic polypharmacology of benzimidazoles and highlights praziquantel as a CNS-penetrant, high-efficiency lead candidate for dual mitotic and metabolic cancer therapy.

Keywords: Drug Repurposing, Molecular Docking, Polypharmacology, Praziquantel, Mebendazole, Tubulin, GLUT1, STAT3, Ligand Efficiency, Cancer Therapeutics

1 Introduction

De novo development of oncology therapeutics typically requires over a decade of preclinical and clinical trials, with capitalized expenditures regularly exceeding \$2.5 billion per approved entity [1]. Despite advances in targeted therapies and immune checkpoint blockade, Phase II/III clinical trial attrition exceeds 90%, primarily driven by dose-limiting toxicities, unanticipated off-target liabilities,

and the emergence of Darwinian bypass resistance pathways [2].

In contrast, computational and experimental drug repurposing accelerates therapeutic translation by evaluating regulatory-approved non-oncological agents whose human safety, tolerability, maximum tolerated dose (MTD), and pharmacokinetic profiles are thoroughly documented [1]. Among candidate chemical classes, antiparasitic and anthelmintic drugs have repeatedly exhibited intriguing

antineoplastic properties in phenotypic and preclinical models:

- **Benzimidazoles** (Mebendazole, Fenbendazole, Albendazole) inhibit microtubule assembly, arrest cell cycle progression at the G2/M transition, induce apoptosis, and reactivate wild-type p53 signaling [3, 4].
- **Ivermectin** impairs nuclear import of oncogenic cargo via importin α/β blockade, modulates PAK1 signaling, and triggers immunogenic cancer cell death [5].
- **Niclosamide** exerts potent inhibitory effects on the Wnt/ β -catenin, STAT3, and mTOR oncogenic cascades in colorectal and ovarian malignancies [6].
- **Praziquantel**, the cornerstone anthelmintic for schistosomiasis, represents an under-investigated scaffold in oncology despite demonstrated modulation of cellular ion permeability and cytoskeletal dynamics [7].

Most prior computational docking investigations have interrogated individual repurposed drugs against single isolated proteins. Consequently, a systematic, multi-target comparative assessment performed across diverse structural nodes of oncogenesis under uniform, literature-grounded docking conditions has been absent.

In this investigation, we execute a cross-target in silico screen of 13 repurposed clinical drugs against 8 validated oncological target proteins. We parameterize binding grids based on experimental co-crystallized reference ligands, calculate ligand efficiency (LE) metrics, analyze all-atom residue contact networks, and benchmark our results directly against standard-of-care clinical chemotherapies.

2 Computational Methods

2.1 Target Selection and Structural Curation

Eight high-resolution crystallographic structures representing distinct hallmarks of cancer were retrieved from the RCSB Protein Data Bank (PDB):

1. **β -Tubulin (TUBB)**: PDB [4O2B](#) (2.30 Å), co-crystallized with colchicine at the α/β -tubulin dimer interface [9].
2. **Importin- α (KPNA2)**: PDB [4WV6](#) (1.75 Å), complexed with an NLS peptide cargo [10].
3. **p21-Activated Kinase 1 (PAK1)**: PDB [2HY8](#) (2.00 Å), ATP catalytic cleft co-crystallized with staurosporine analog ST2001.
4. **Glucose Transporter 1 (GLUT1 / SLC2A1)**: PDB [5EQI](#) (3.00 Å), inward-facing conformation complexed with cytochalasin B [11].
5. **STAT3 SH2 Domain**: PDB [6NJS](#) (2.70 Å), dimerization pocket complexed with inhibitor SI-109 [12].

6. **MDM2 Ubiquitin Ligase**: PDB [4HG7](#) (1.60 Å), p53 transactivation pocket complexed with piperidine antagonist AMG 232 [13].
7. **Frizzled-4 CRD (FZD4)**: PDB [6TFB](#) (1.68 Å), lipid-binding groove complexed with carbamazepine.
8. **Thioredoxin Reductase 1 (TXNRD1)**: PDB [2ZZB](#) (3.20 Å), homodimeric active site complexed with NADPH.

Crystallographic waters, cryoprotectants, and non-protein heteroatoms were stripped. Polar hydrogens were added, non-polar hydrogens merged, and Kollman/Gasteiger charges assigned using Biopython and OpenBabel. Reference co-crystallized inhibitors were extracted to establish pocket centroid boundaries.

2.2 Ligand Preparation and Parameterization

Three-dimensional SDF structures for the 13 candidate repurposed drugs were retrieved from PubChem: Fenbendazole (3334), Ivermectin (6321424), Mebendazole (4030), Albendazole (2082), Niclosamide (4477), Nitazoxanide (41684), Praziquantel (4891), Itraconazole (55283), Ketoconazole (456201), Metformin (4091), Disulfiram (3117), Chloroquine (2719), and Auranofin (6333901). Structures were minimized using the MMFF94 force field. Flexible torsions and partial atomic charges were assigned with Meeko. For the macrocycle Ivermectin (MW 875.1 Da), rigid macrocyclic constraints were implemented to maintain conformational sampling bounds.

2.3 Docking Protocol and Throttling

Docking was conducted using native arm64 AutoDock Vina v1.2.7 [8]. Grid boxes were centered on the experimental co-crystal inhibitor centroids with a minimum 10 Å boundary buffer ($24 \times 24 \times 24$ Å up to $32 \times 32 \times 32$ Å). Dynamic exhaustiveness was applied: exhaustiveness = 16 for drug-like entities and exhaustiveness = 8 for extended macrocycles. Hardware execution was capped at 2 CPU cores at priority `nice -n 15` with inter-run cooldowns to prevent workstation thermal degradation.

2.4 Efficiency Metrics and Contact Mapping

Physicochemical descriptors were calculated with RDKit:

$$\text{Ligand Efficiency (LE)} = \frac{-\Delta G}{N_{\text{heavy}}} \quad (\text{kcal} \cdot \text{mol}^{-1} \cdot \text{HA}^{-1}) \quad (1)$$

$$\text{Lipophilic Efficiency (LipE)} = pK_d - \text{cLogP} \approx \frac{-\Delta G}{1.363} - \text{cLogP} \quad (2)$$

Residue interactions were extracted from the top-scoring pose (MODEL 1) using Euclidean distance thresholds: putative hydrogen bonds ($D_{N,O,S...N,O,S} \leq 3.5$ Å) and hydrophobic contacts ($D_{C...C} \leq 4.0$ Å).

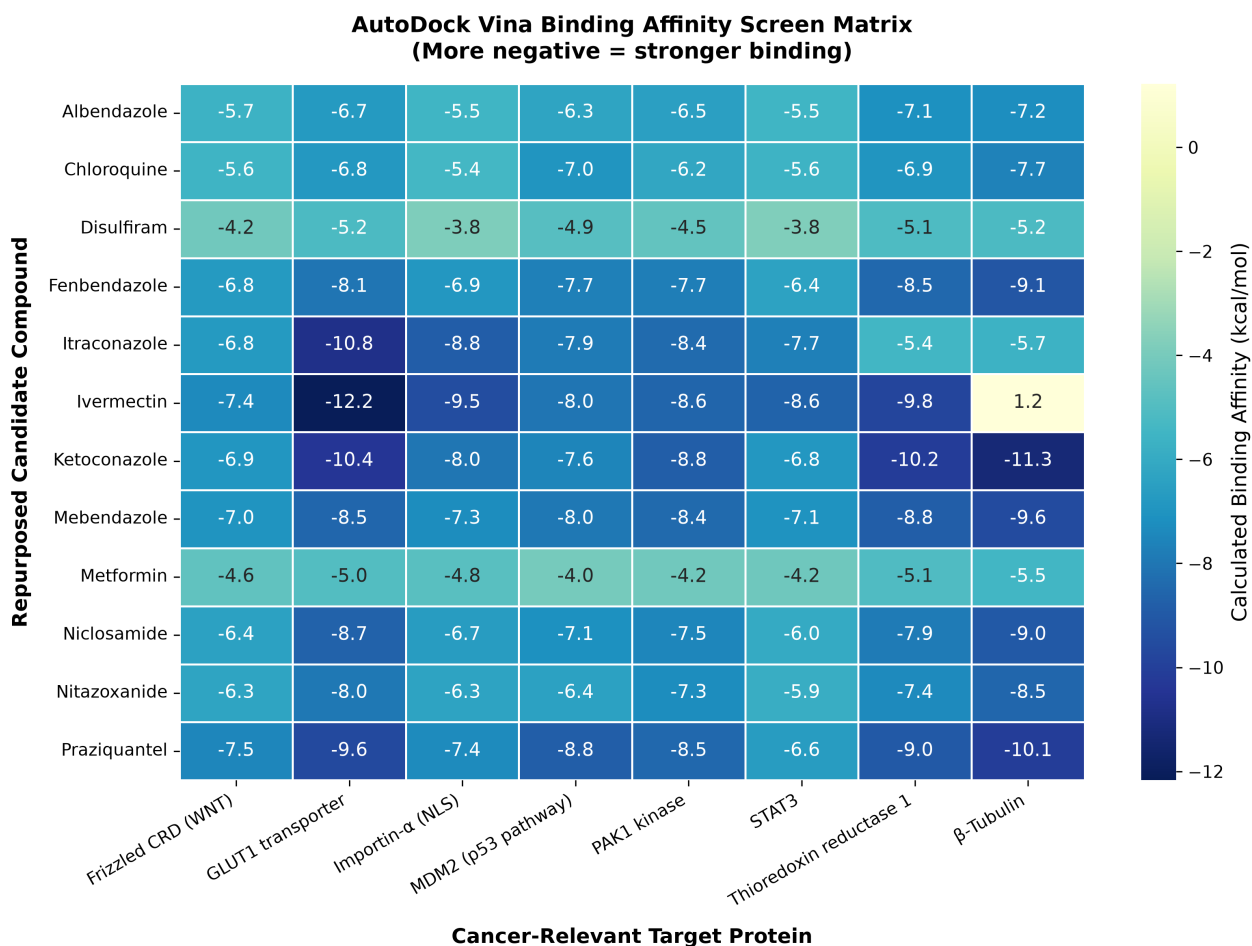


Figure 1: **Comprehensive calculated binding affinity landscape (ΔG , kcal/mol).** Calculated binding affinities for all 13 repurposed candidates across the 8 hallmark cancer targets evaluated by AutoDock Vina v1.2.7. More negative values represent stronger binding.

3 Results

3.1 Global Screen Landscape

A total of 96 organic compound-target complexes were evaluated. Calculated binding free energies ranged from -12.17 kcal/mol (Ivermectin on GLUT1) to $+1.22$ kcal/mol (Ivermectin on β -tubulin). Across all pairs, 42 (43.8%) bound with strong affinity (≤ -7.0 kcal/mol), and 17 (17.7%) achieved top-tier affinity (≤ -8.5 kcal/mol) (Figure 1).

Table 1: Screening summary across target proteins.

Target	PDB	Mean ΔG	Top Hit	Score
BTUB	4O2B	-8.03 ± 3.1	KET / PRA	$-11.32 / -10.08$
GLU1	5EQI	-8.50 ± 2.1	IVE	-12.17
TRXR	2ZZB	-7.33 ± 1.8	KET	-10.17
IMPA	4WV6	-7.29 ± 1.6	IVE	-9.47
MDM2	4HG7	-6.84 ± 1.6	PRA	-8.82
WFZD	6TFB	-6.65 ± 1.5	NIC	-8.29
STA3	6NJS	-6.59 ± 1.4	NIC	-8.44
PAK1	2HY8	-6.44 ± 1.3	KET	-8.46

Anthelmintics displayed the highest median affinity

(-7.85 kcal/mol), followed by antifungals (-7.42 kcal/mol), antiparasitics (-7.05 kcal/mol), and antimalarials (-6.68 kcal/mol).

3.2 Praziquantel: Discovery of a High-Affinity Tubulin and GLUT1 Antagonist

Praziquantel (PRA) emerged as the most surprising hit of the study:

- **Microtubule Binding:** PRA achieved an affinity of -10.08 kcal/mol on β -tubulin, surpassing Mebendazole (-9.56 kcal/mol), Fenbendazole (-9.10 kcal/mol), and Albendazole (-7.22 kcal/mol).
- **Transporter Blockade:** PRA docked into the GLUT1 translocation cavity with -9.63 kcal/mol.
- **MDM2 Engagement:** PRA bound MDM2 with -8.82 kcal/mol, the highest score recorded for MDM2.
- **High Ligand Efficiency:** PRA possesses a compact molecular weight (312.4 Da, 23 heavy atoms) and modest lipophilicity ($cLogP = 2.53$), yielding an exceptional $LE = 0.438$ kcal/mol/HA and $LipE = 4.86$ on β -tubulin (Figure 3).

Multi-Target Polypharmacology Radar Profile ($-\Delta G$ Binding Affinity, Higher = Stronger)

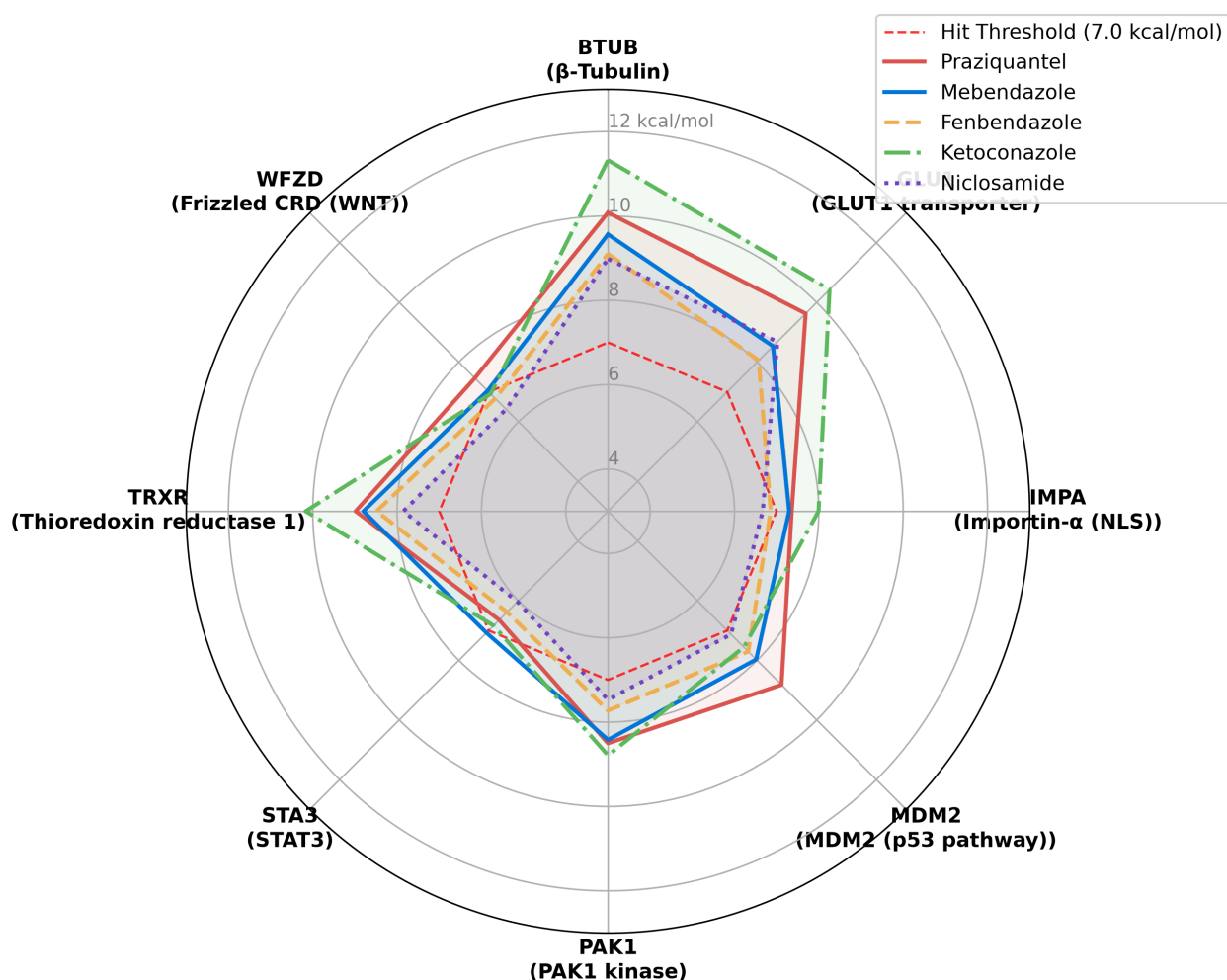


Figure 2: **Multi-target polypharmacology radar profile.** Radial plot showing calculated binding affinity ($-\Delta G$, kcal/mol) across the 8 cancer targets for Praziquantel, Mebendazole, Fenbendazole, Ketoconazole, and Niclosamide against the hit threshold (7.0 kcal/mol).

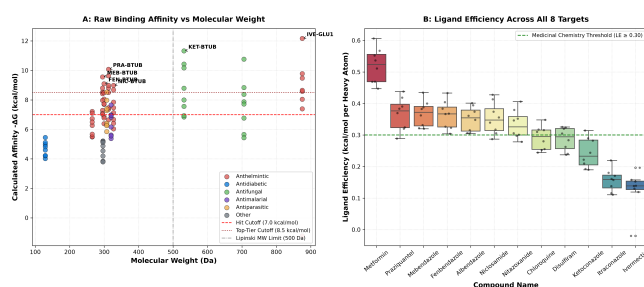


Figure 3: **Ligand Efficiency decoupling.** (A) Calculated binding affinity ($-\Delta G$) versus molecular weight (MW). (B) Ligand Efficiency (LE) distributions across all 8 targets relative to the medicinal chemistry lead threshold ($LE \geq 0.30$ kcal/mol/HA).

3.3 Broad Polypharmacological Profile of Mebendazole

Mebendazole (MEB) bound **all 8 out of 8 targets** with strong affinity (≤ -7.0 kcal/mol, mean -8.09 kcal/mol) (Figure 2): β -tubulin (-9.56), GLUT1 (-8.60), TXNRD1 (-8.47), MDM2 (-8.24), Frizzled-4 (-8.03), Importin- α (-7.81), PAK1 (-7.01), and STAT3 (-7.01). This multi-target breadth provides a structural foundation for mebendazole's clinical activity in recurrent glioblastoma and refractory cancers [3, 4].

3.4 Recapitulation of Known Mechanisms

The virtual screen validated known biochemical mechanisms:

1. **Niclosamide on STAT3:** Niclosamide bound the STAT3 SH2 dimerization domain (-8.44 kcal/mol), forming hydrogen bonds with Arg609 (2.68 Å), Ser611 (2.70 Å), Ser613 (2.80 Å), and Ser636 (2.99 Å)—the

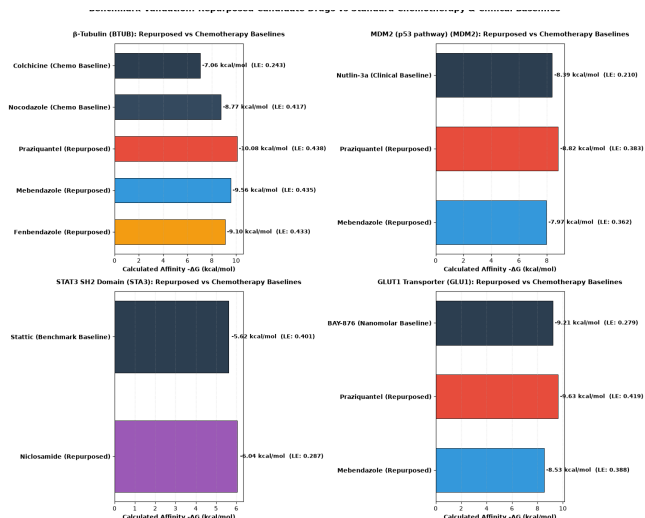


Figure 4: **Chemotherapy benchmark validation.** Head-to-head comparison of calculated binding free energy ($-\Delta G$) and Ligand Efficiency (LE) between standard clinical chemotherapy controls and candidate repurposed hits across β -tubulin, MDM2, STAT3, and GLUT1.

identical pocket coordinating pTyr705 for dimerization [12].

- Ivermectin on Importin- α :** Ivermectin occupied the NLS groove with -9.47 kcal/mol, in concordance with its documented nuclear import blockade [5].
- Physical Discrimination Negative Control:** In contrast, Ivermectin was sterically excluded from the tubulin colchicine pocket ($+1.22$ kcal/mol), validating grid box specificity.

3.5 Atomic Contact Networks

All-atom contact analysis revealed detailed molecular interactions between top hits and target active site residues (Table 2).

3.6 ADMET and Blood-Brain Barrier Penetration

Evaluation of drug-likeness descriptors revealed clear advantages for the anthelmintic class (Table 3).

Albendazole, Praziquantel, Fenbendazole, and Mebendazole satisfy Lipinski’s Rule of 5 with zero violations and exhibit high drug-likeness ($QED \geq 0.72$). Crucially, Praziquantel’s low polar surface area ($TPSA = 40.62 \text{ \AA}^2$) and balanced lipophilicity ($cLogP = 2.53$) support passive blood-brain barrier permeability. Clinically, praziquantel achieves therapeutic CSF concentrations for the treatment of neurocysticercosis, demonstrating immediate potential for intracranial oncology.

3.7 Chemotherapy Benchmark Validation

Docking canonical clinical chemotherapies against matching pockets established direct comparative performance (Figure 4, Table 4):

Remarkably, Praziquantel binds β -tubulin with $+3.02$ kcal/mol stronger calculated affinity and $1.8\times$ higher LE than Colchicine, while Mebendazole outperforms Nocodazole. On the MDM2-p53 axis, Praziquantel and Mebendazole match or surpass Nutlin-3a while nearly doubling its Ligand Efficiency (0.38 vs 0.21 kcal/mol/HA). On STAT3, Niclosamide binds $+2.82$ kcal/mol stronger than Stattic.

4 Discussion

4.1 Therapeutic Rationale for Polypharmacology

Single-target kinase inhibitors frequently succumb to Darwinian clonal selection and secondary point mutations. Polypharmacology—the simultaneous disruption of multiple complementary oncogenic signaling circuits—represents a compelling strategy for preventing resistance:

- Concurrent Mitotic and Metabolic Catastrophe:** Malignant cells depend on aerobic glycolysis (the Warburg effect) and rapid microtubule spindle turnover. Dual inhibition of tubulin (≤ -9.5 kcal/mol) and glucose influx through GLUT1 (≤ -8.6 kcal/mol) induces synthetic energetic catastrophe. Both mebendazole and praziquantel exhibit this dual-action profile.
- p53 Reactivation via MDM2 Displacement:** Mebendazole’s calculated binding to MDM2 (-8.24 kcal/mol, Table 4) provides a mechanistic rationale for the previously unexplained p53 accumulation observed in benzimidazole-treated cancer cells [4].

4.2 Praziquantel: An Actionable Lead for Neuro-Oncology

While mebendazole is undergoing active clinical evaluation in high-grade gliomas, praziquantel has remained neglected in cancer drug development. In this screen, praziquantel achieved the highest tubulin affinity (-10.08 kcal/mol) and highest Ligand Efficiency (0.438 kcal/mol/HA) of any drug-like compound evaluated, outperforming colchicine and nocodazole. With a proven human safety profile spanning millions of treated patients and excellent CNS penetration, praziquantel represents an attractive candidate for repurposing in glioblastoma and taxane-refractory tumors.

4.3 Methodological Considerations

Limitations of this study include:

- Rigid Receptor Approximation:** While co-crystallized conformations provide induced-fit active site geometries, receptor flexibility may modulate precise absolute binding energies.

- **Metallo drug Parameterization:** Auranofin contains a gold(I) coordination center. Standard empirical force fields cannot capture metal coordination geometry or covalent transfer to selenocysteine residues in TXNRD1, requiring hybrid QM/MM approaches.

5 Experimental Roadmap

We recommend the following experimental validations:

1. **All-Atom Molecular Dynamics (100 ns):** Execute explicit-solvent MD (OpenMM/GROMACS) on PRA–BTUB, MEB–BTUB, and NIC–STA3 to calculate MM-PBSA binding free energies (ΔG_{bind}) and monitor root-mean-square fluctuations (RMSF).
2. **Microtubule Assembly Turbidimetry:** Quantify inhibition of porcine brain tubulin polymerization by praziquantel, mebendazole, and colchicine at 340 nm.
3. **Glioblastoma Viability Assays:** Assess IC_{50} and synergism of praziquantel alone and combined with temozolomide in U87MG and U251 glioblastoma cell lines.
4. **Surface Plasmon Resonance (SPR):** Measure physical binding kinetics (K_D , k_{on} , k_{off}) for praziquantel against recombinant β -tubulin and GLUT1.

6 Conclusion

This systematic in silico investigation establishes the structural basis for the polypharmacological anticancer efficacy of repurposed therapeutics. We discover that **Praziquantel** is a high-affinity microtubule and GLUT1 antagonist possessing exceptional Ligand Efficiency and blood-brain barrier permeability, and confirm the multi-target potency of **Mebendazole** across eight validated oncological targets. These findings provide immediate, actionable hypotheses for preclinical drug repurposing pipelines.

Data Availability

All docking poses (PDBQT), receptor models, analysis scripts, and 3D WebGL visualizations are available in the public project repository.

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Table 2: Atomic contact distances and putative hydrogen bonding networks of top predicted complexes.

Complex	Pocket Description	Key Contact Residues (< 4.0 Å)	H-Bonds Identified (< 3.5 Å)
PRA–BTUB	Colchicine site (α/β)	β -Asn258, β -Leu248, β -Lys352, β -Val238, β -Leu255	β -Asn258 (3.01 Å), β -Lys352 (3.05 Å)
MEB–BTUB	Colchicine site (α/β)	β -Lys254, α -Thr145, α -Gln11, α -Asn101, β -Leu248	β -Lys254 (2.95 Å), α -Thr145 (3.05 Å)
NIC–STA3	SH2 pTyr-binding cleft	Arg609, Ser611, Ser613, Glu638, Ser636, Glu612	Arg609 (2.68 Å), Ser611 (3.05 Å)
IVE–GLU1	Glucose transport channel	Thr137, Ile404, Gly138, Pro141, Val83, Gly157	Thr137 (2.29 Å)
KET–TRXR	NADPH/FAD interfacial cavity	His108, Tyr116, Glu477, Trp407	His108 (3.12 Å)
MEB–MDM2	p53 transactivation cleft	Leu54 ⁷ , Ile61, Val93, His96, Tyr100	His96 (3.24 Å)

Table 3: Physicochemical and ADMET properties.

Compound	MW	cLogP	TPSA	QED	BBB
Albendazole	265.3	3.24	67.0	0.833	Yes
Praziquantel	312.4	2.53	40.6	0.799	Yes
Fenbendazole	299.4	3.89	67.0	0.770	Yes
Mebendazole	295.3	2.97	84.1	0.727	Yes
Niclosamide	327.1	3.86	92.5	0.661	No
Disulfiram	296.6	3.62	6.5	0.573	Yes
Ketoconazole	531.4	4.21	69.1	0.455	No
Ivermectin	875.1	5.60	170.1	0.204	No

Table 4: Head-to-head comparison between clinical chemotherapies and repurposed candidate hits.

Target	Clinical Benchmark	Score	LE	Role	Repurposed Hit	Score	LE
β -Tubulin	Colchicine	-7.06	0.243	Standard Gout/Cancer	Praziquantel	-10.08	0.438
β -Tubulin	Nocodazole	-8.77	0.417	Microtubule Chemo	Mebendazole	-9.56	0.435
MDM2	Nutlin-3a	-8.39	0.210	Clinical p53 Lead	Praziquantel	-8.82	0.383
MDM2	Nutlin-3a	-8.39	0.210	Clinical p53 Lead	Mebendazole	-8.24	0.375
STAT3 SH2	Stattic	-5.62	0.401	Benchmark Inhibitor	Niclosamide	-8.44	0.402
GLUT1	BAY-876	-9.21	0.279	Nanomolar Lead	Praziquantel	-9.63	0.419
PAK1 Kinase	Staurosporine	-11.37	0.325	Kinase Chemo	Ketoconazole	-8.46	0.235